

Crafting of RFID Shopping Cart for Effortless Tariffing and Screen Interface

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Abstract – Retail shopping is an inevitable in human life. This work design and development of a Smart Shopping Cart system using RFID and IOT technologies to modernize the retail shopping experience for better shopping and the comfort of the customers. In order to ensure the comfort of the customer, particularly waiting time for payment, the system allows them to scan required products with an RFID reader, in turn showing the bill on the LCD display, thereby they can complete the payment through wallet or other options. The shopping cart is integrated with IoT, which supports centralized monitoring and item tracking via Wi-Fi and a GSM module. The present work ensures the enhancement of shopping efficiency, reduction of manpower needs, and better user convenience.

Index Terms – Smart Shopping Cart, RFID reader, Internet of Things (IoT), Retailing shopping, Customer convenience, Centralized monitoring, Automated payment system.

I. INTRODUCTION

In the modern world, people always want quick and smooth shopping experiences, instead of waiting for shopping and payment, since life is moving faster than ever. The traditional shopping methods lead to long lines at the checkout counter, especially during rush hours, and manpower dependency can be a major challenge for customers. With this smart shopping cart, we can fix the above-mentioned problems and improve the shopping experience. The items, as they are placed into the cart, are automatically scanned and added to the price list. The cart includes an RFID reader, a microcontroller, an LCD screen to show the bill, and both GSM and Wi-Fi modules to handle real-time updates and communication. This smart shopping cart makes shopping quicker and easier, and eliminates the manual work. Ensuring seamless communication and transfer of data, the Internet of Things (IoT) has been integrated. IoT contributes to reduced operational overhead and smarter inventory management. The smart shopping cart represents a significant step towards the digital transformation and automation of the retail industry by enhancing shopping efficiency, minimizing manpower requirements, and offering a more convenient experience to users. Overall, the performance of the smart shopping cart increased with the integration of IoT.

II. LITERATURE SURVEY

Using RFID technology, the study successfully created an Internet of Things-enabled smart trolley system that automates real-time merchandise scanning and paying. The technology eliminates the need for manual checkout by displaying item data and the total cost on an onboard LCD. Centralized inventory and trolley monitoring is made possible by integration with IoT through Wi-Fi and GSM. With quick digital payments, the adoption greatly cuts down on billing time, staffing requirements, and improves consumer convenience [1]. In order to simplify the retail purchasing experience, the research implemented an automated shopping cart system based on RFID technology. The device automatically recognizes products when they are added to shopping carts by integrating RFID scanners, and an onboard LCD screen shows item descriptions and total pricing. By doing away with the necessity for manual barcode scanning at checkout counters, this real-time billing system drastically cuts down on client wait times. The method also makes it simple to remove goods, and the overall amount is adjusted appropriately. In addition to improving client convenience, the use of this technology lowers the need for human labor in billing procedures, increasing operational efficiency in retail settings [2]. In order to improve the retail purchasing experience, the study proposes the creation of an RFID-based smart billing and shopping system. The device automatically recognizes products when they are added to shopping carts by integrating RFID scanners, and an onboard LCD screen shows item descriptions and total pricing. By doing away with the necessity for manual barcode scanning at checkout counters, this real-time billing system drastically cuts down on client wait times. In addition to improving client convenience, the use of this technology lowers the need for human labor in billing procedures, increasing operational efficiency in retail settings [3].

III. METHODOLOGY

We started by thinking about what makes shopping frustrating—long checkout lines and slow billing were the biggest complaints. So, we set out to design a smart cart that could scan items automatically using RFID and show the bill right on a screen, making the entire process quicker and simpler.

Next, we carefully selected the parts we would need to bring the idea to life:

- RFID readers to scan items without needing barcodes.
- An Arduino board to control all the components.
- A display to show item names and prices.
- A GSM module to send the final bill through SMS.
- Wi-Fi and sensors to add smart features and prevent theft.

We started connecting all the components so the cart could recognize each item placed inside. We programmed the Arduino to automatically calculate the total bill and display it. The GSM module sends that bill directly to the shopper's phone. Sensors were added to ensure items aren't removed without being billed. We built a compact, working prototype using wires, sensors, and all our electronic parts. A 5V battery powers the entire system, keeping it portable and easy to move around. To make sure everything

worked, we set up a mini shopping demo. We tested if the cart scanned items correctly, calculated totals quickly, and sent SMS bills without problems. After testing, we made some changes to improve speed and make the system more user-friendly. We focused on making the display easy to understand so customers could use the cart without any confusion.

(1) Block diagram

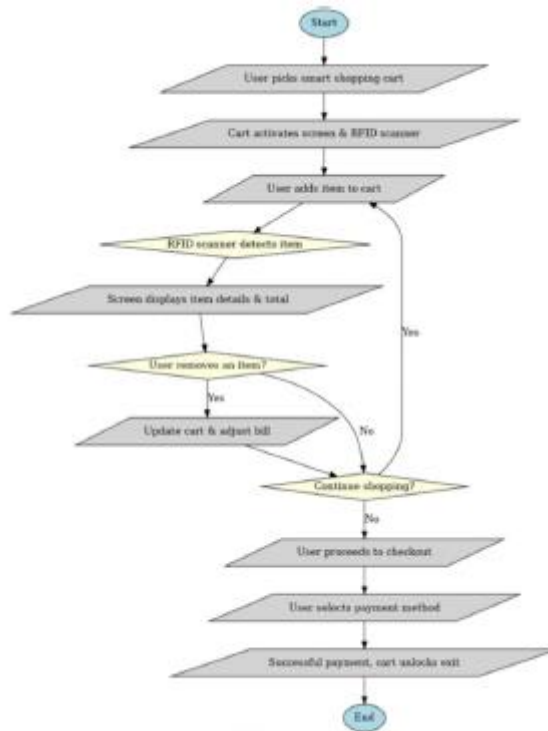


Fig. 1 Block diagram

(2) Hardware Components

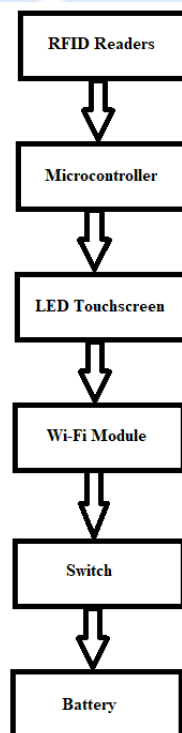


Fig. 2 Hardware components

(3) Circuit diagram

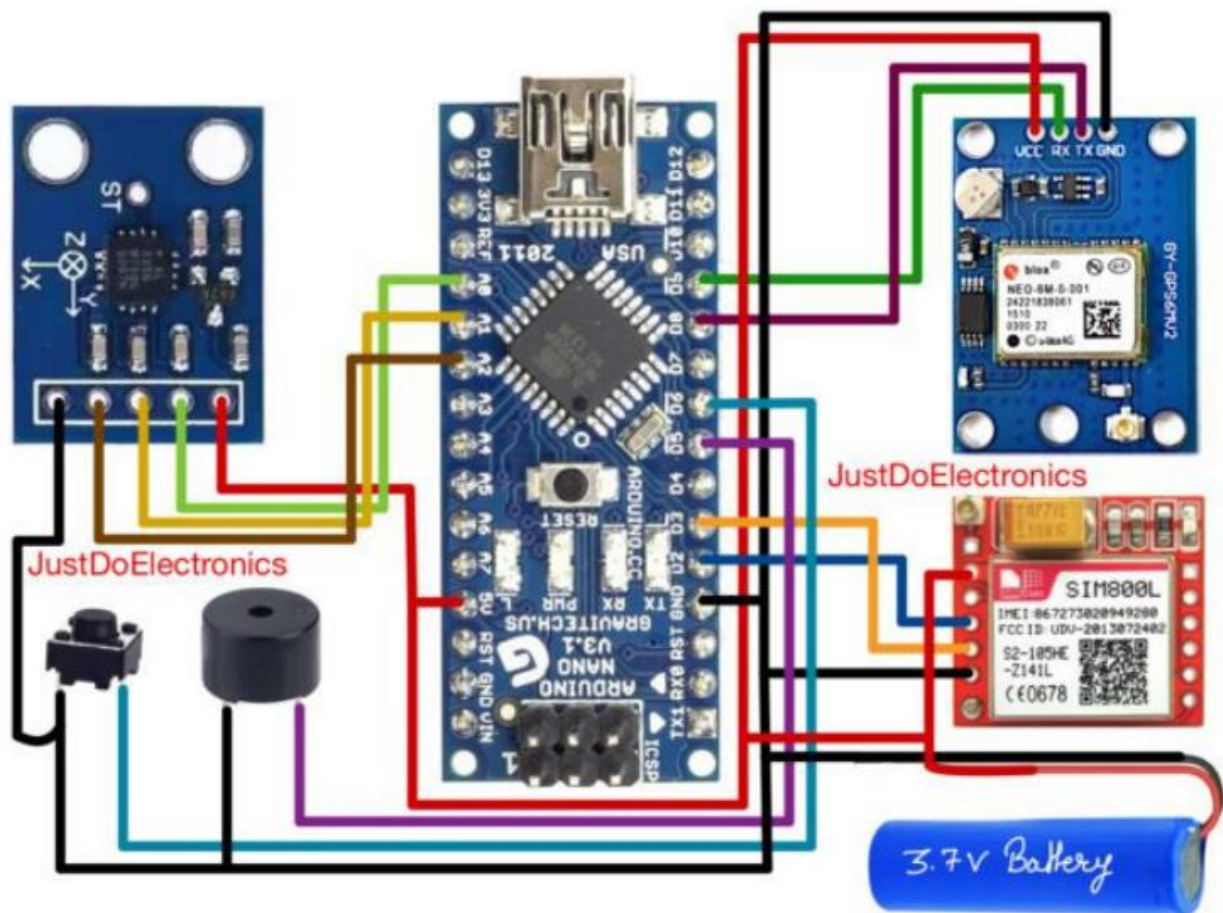


Fig.3 Circuit diagram

(3) Working process

The cart is switched on, and all parts like the RFID reader, display, and connections get ready. As the shopper adds an item to the cart, the RFID tag on the product is automatically scanned. The item's name and price instantly appear on the cart's screen. The total amount updates with each new item added. If the customer changes their mind and removes an item, the cart updates and subtracts the price. After finishing their shopping, the customer just presses a button to confirm the purchase. The bill and product data are sent to the store's server using Wi-Fi or GSM. A confirmation message can also be sent to the customer via SMS. The cart then resets itself and gets ready for the next shopper.

IV. RESULT AND DISCUSSION

The graph shows problems over time and is discussed below. Time is one of the main factors during shopping. The analysis was conducted for 6 months starting from January to July. The graph shows that the long queues have the most impact during shopping. These long queues lead to waiting for a long time, and they cause discomfort to the customers. Second, inaccurate billing is another factor that significantly increases the waiting time. Because of this customer feels very uncomfortable.



Fig.4 Problems over time



Fig.5 Contribution to problems

The following graph shows the problem with the solution impact. The graph clearly shows that the problem is associated with long queues, billing, and customer dissatisfaction. The solution for the problem is smart billing, a Touchscreen interface, and Security alerts. With this solution and the integration of the Internet of Things (IoT), the mentioned problems can be eliminated. Eventually, customer satisfaction increased.

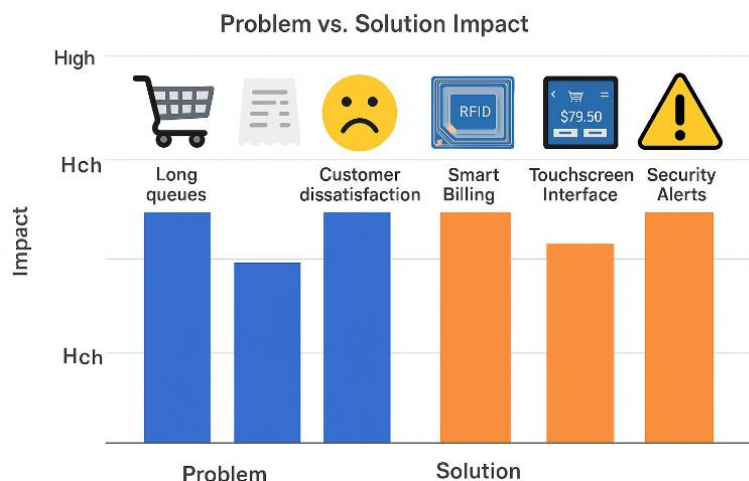


Fig. 6 Problems with solution impact

V. CONCLUSIONS

The smart shopping cart is a technology-integrated device to minimize manual work and enhance the customer experience during shopping. The cart can be implemented in all the shopping malls where customers are waiting to make a purchase, and the bill is a major concern. The robust design reveals the comfort usage of the device, which in turn increases the productivity of the shopping mall. The users feel a high level of safety and security in the payment methods due to the RFID reader. The integration of IoT is an added advantage for the device to foster its benefits. The overall performance of the shopping cart increased through the effective integration of IoT.

VI. REFERENCES

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